Global CO_2 Emissions:

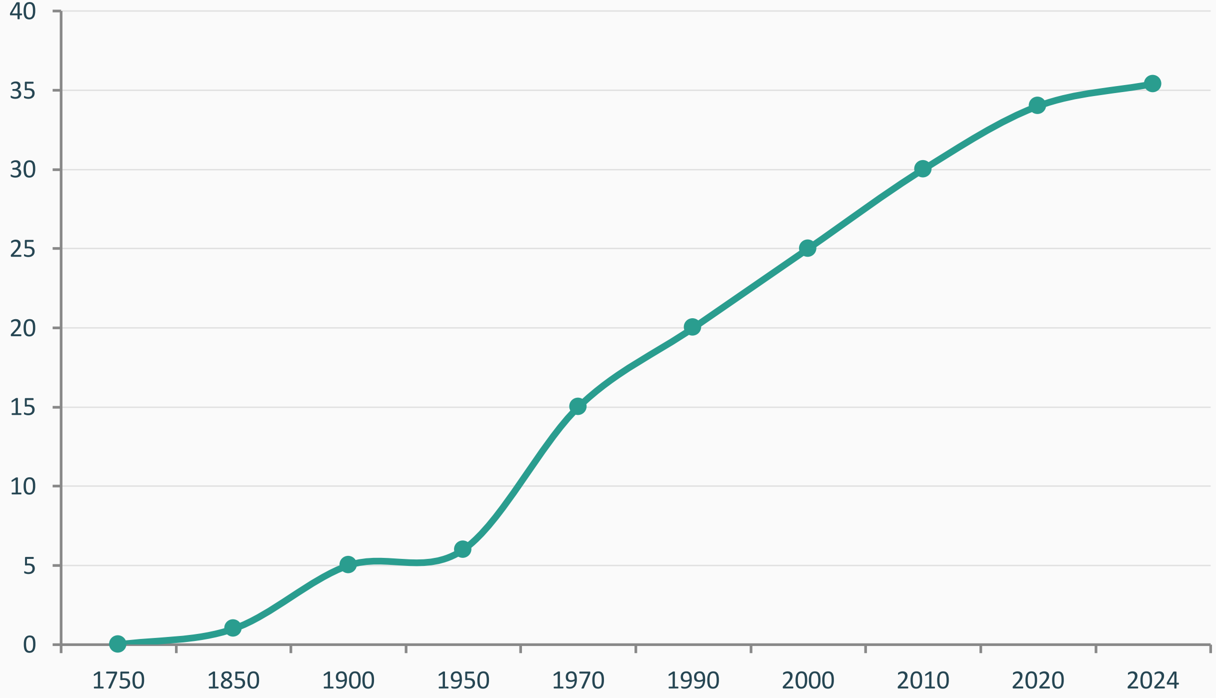
Who Emits, How Much, and What's Changing

Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data · Hannah Ritchie & Max Roser

For policy analysts, climate negotiators, and sustainability researchers

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Industrial Revolution

Coal power began driving emissions from ~1750

Post-WWII Acceleration

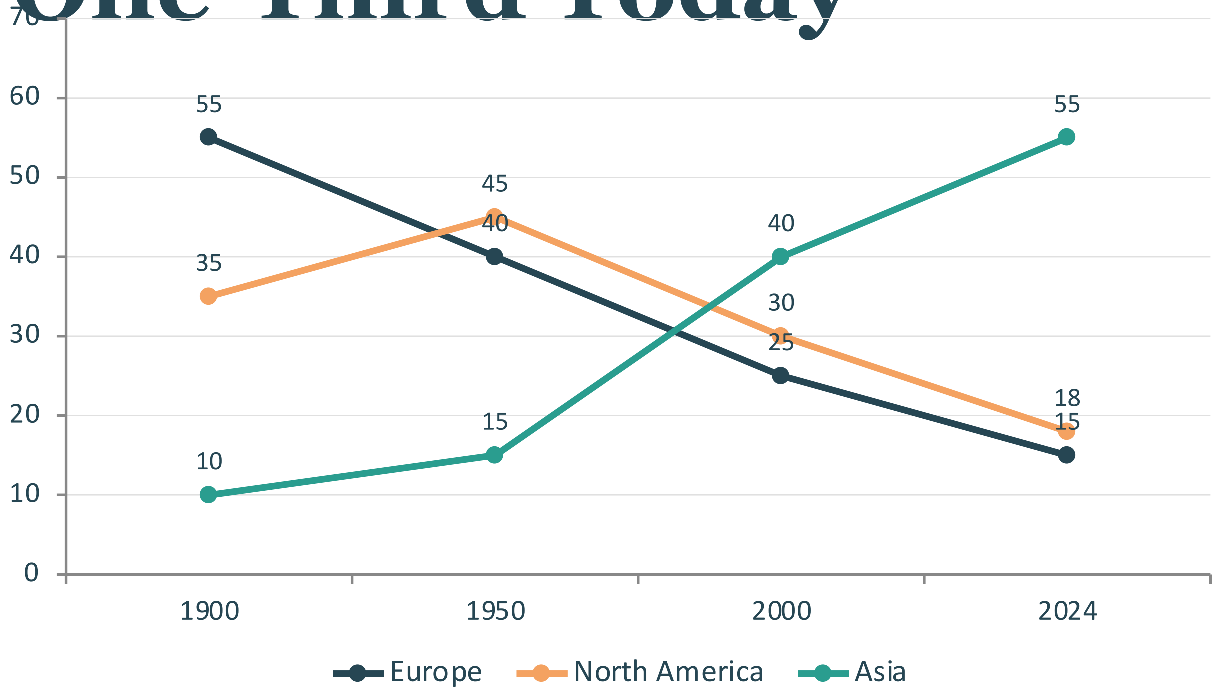
Rapid industrialization pushed emissions from 6 to 15 Bt in just 20 years

Growth Slowing, No Peak Yet

Despite climate efforts, emissions continue rising toward 35+ Bt in 2024

Source: Global Carbon Budget 2025 via Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



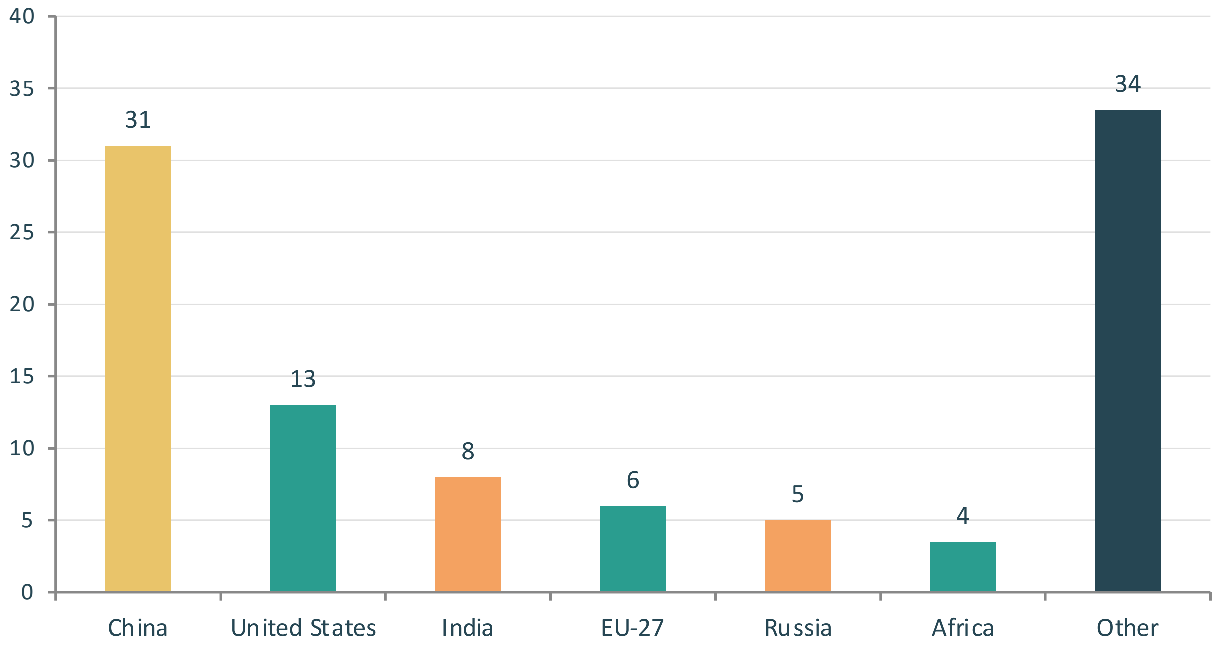
Asia now produces over half of global emissions

Europe + NA share fell from 90% to under 33%

Source: Global Carbon Project, 2024

China Now Emits More Than One-Quarter of Global CO_2

Share of global CO_2 emissions by country/region (2024)



31%

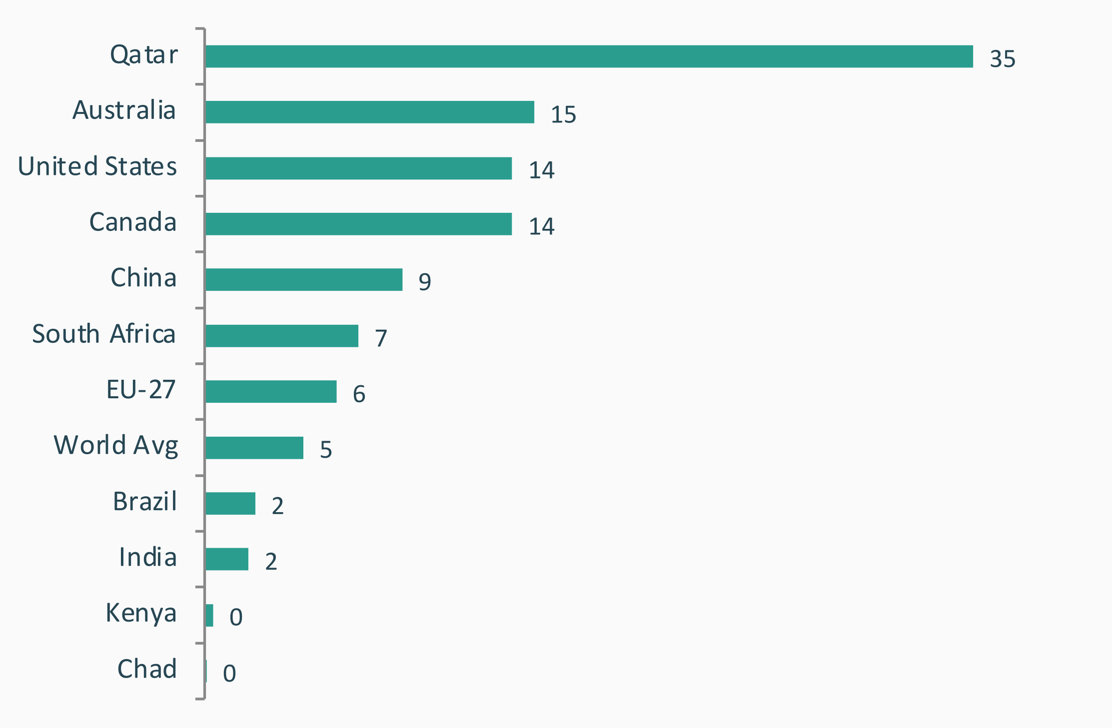
China's Share

>25%

of global emissions

Source: Global Carbon Project, 2024

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150x

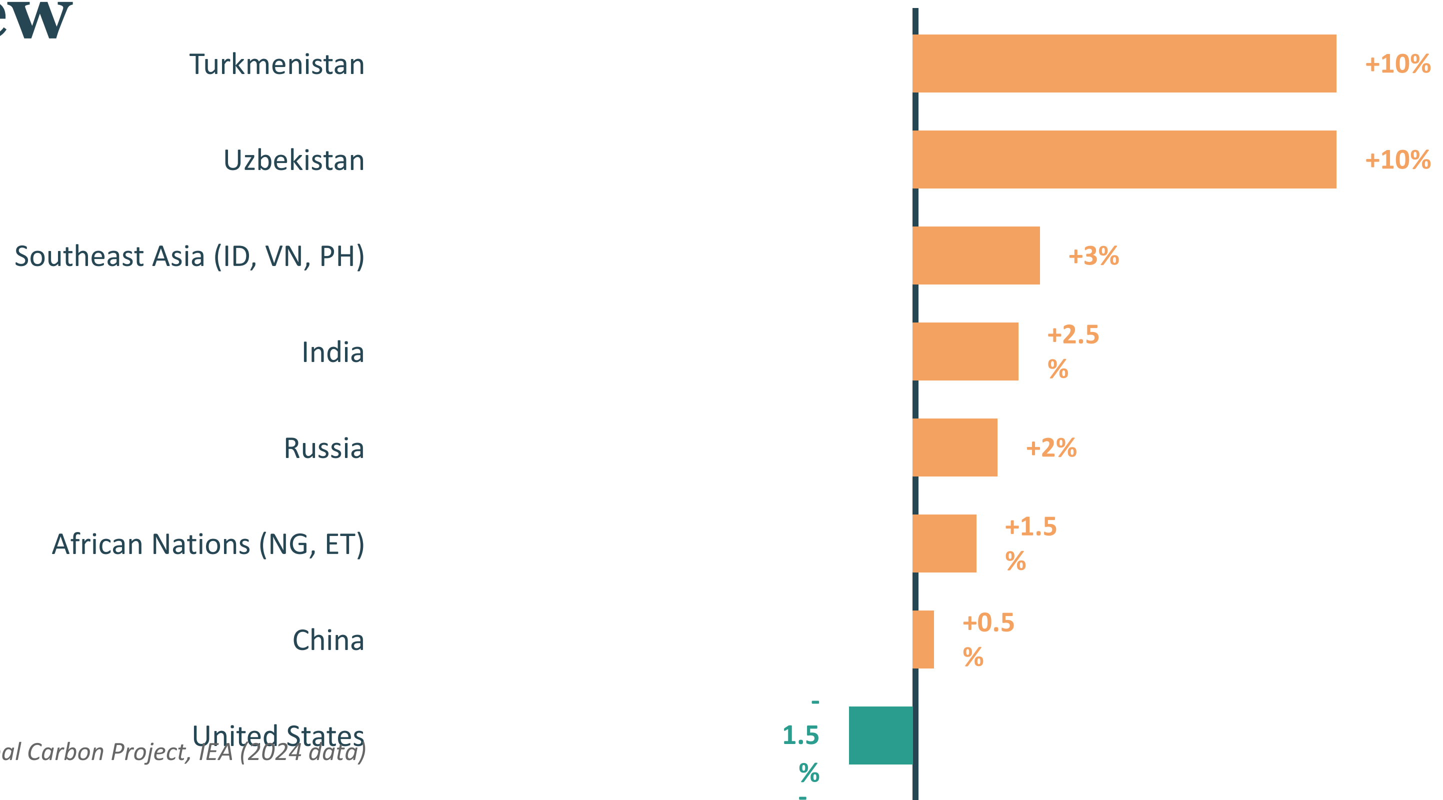
gap between lowest and highest emitters

Per capita emissions差距显著

Source: Our World in Data, Global Carbon Project (2023)

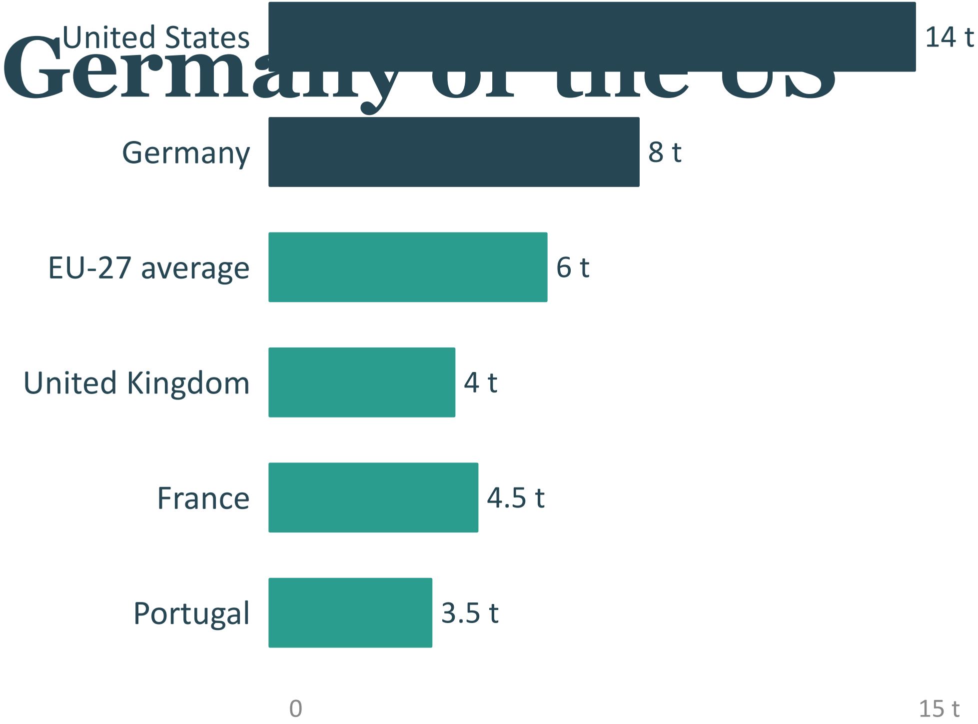
2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual % Change in CO2 Emissions by Country/Region



Source: Global Carbon Project, IEA (2024 data)

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Source: IEA, Our World in Data (2024)

LOW-EMISSION CLUB

France (4.5 t) and Portugal (3.5 t) have among the lowest per capita emissions among wealthy nations — thanks to nuclear and renewable-heavy energy mixes.

3x DIFFERENCE

US (14 t per capita) vs France (4.5 t per capita)

Similar prosperity, very different energy choices.

Among wealthy nations with similar prosperity levels, per capita emissions vary significantly because of energy mix choices. France, the UK, and Portugal have much lower per capita emissions because a higher share of their electricity comes from nuclear and renewable sources.

Key Takeaways:
Responsibility Is Shared but Unequal

- Global emissions exceed 35 Bt/yr and haven't peaked
- China is the largest annual emitter but per capita still below the US
- Historical responsibility is dominated by Europe and the US
- Per capita gaps of 150x persist between richest and poorest nations
- Energy policy and technology choices can decouple prosperity from emissions

Source: Global Carbon Budget 2025 via Our World in Data